

Memory, Addressing and Pointers

Introduction to ARM Cortex-M Microprocessors – Chapter 3

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Chapter 2 ended with a question. A 4 GB address space, filled with registers, peripherals, and flags, is an architectural fact. But what does this actually mean for a C program?

A microcontroller has no instruction that means "turn on the LED." It only has instructions that read and write numbered locations. So controlling hardware becomes one simple question: *which number do I write, and to which address?* This chapter builds the bridge between the C you already know and the addresses the processor actually uses. It is the shortest path from `int x;` to `GPIOA->ODR`. Everything in Parts III and IV depends on it.

After completing this chapter, you should be able to:

1. Explain what an address is and how it differs from the value stored at that address.
2. Use the `&` and `*` operators to obtain an address and to access the value it points to.
3. Express the size of a byte, halfword, and word on a 32-bit ARM processor, and compute the address of any element of an array.
4. Select the correct fixed-width type (`uint8_t`, `uint16_t`, `uint32_t`) for a given range of data.
5. Explain the effect of `const` and `volatile` on how the compiler generates memory accesses.
6. Write a C pointer that reads and writes a specific hardware register given its physical address.

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Memory is organized as a sequence of one-byte locations. Each byte has a unique **address**, which acts as its location number.

A 32-bit address can identify

$$2^{32} = 4,294,967,296$$

different byte locations.

Address	<i>where</i> a byte is located
Contents	<i>what</i> is stored at that location

Keep these two apart

A house number and the family living in it are different things. An address identifies a location; it is not the value stored there.

A 32-bit word is stored in memory starting at 0x2000 looks like this:

<i>Address</i>	<i>Contents</i>
0x2000	0xDE
0x2001	0xC0
0x2002	0xAD
0x2003	0xDE

Notice that the addresses increase by one because memory is **byte-addressable**: each address identifies one byte.

A larger value occupies several consecutive byte locations. For example, a 32-bit word occupies four bytes.

So, an address identifies the *first byte* of a multi-byte object. The remaining bytes follow at the next addresses.

Suppose `uint32_t x = 42;` and the compiler puts `x` at `0x20000100`. We can ask two different questions:

<i>Where does <code>x</code> live?</i>	<code>0x20000100</code> — the address
<i>What value does <code>x</code> contain?</i>	<code>42</code> — the value

In C, `x` names the value, while `&x` produces its address.

You already know this from digital electronics

An address is the input to a **decoder**. With memory-mapped I/O, the decoder can select peripheral registers as well as RAM locations.

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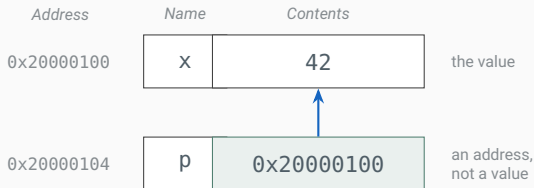
A Pointer Holds an Address A variable whose value is an address

We just saw that `&x` gives the address where `x` lives. C needs somewhere to **keep** that address. A **pointer** is a variable built for exactly this job.

- `&x` — **address-of**: the address at which `x` is stored
- `*p` — **dereference**: the value stored at the address held in `p`

```
uint32_t x = 42;
uint32_t *p = &x; // p holds x's
                  // ADDRESS
uint32_t v = *p;   // v becomes 42
*p = 99;           // x becomes 99
```

These two undo each other: `*(&x)` is just `x`.





Address	Contents
0x20000100	readings[0]
0x20000104	readings[1]
0x20000108	readings[2]
0x2000010C	readings[3]

An array is a block of **elements**, placed one after another, all the same size. So its addresses follow a formula.

$$\text{address of element } i = \text{base} + (i \times \text{element size})$$

For readings[2]: $0x20000100 + (2 \times 4) = \mathbf{0x20000108}$.

The processor can perform this scaling as part of the memory access instruction.

Pointer Arithmetic

A pointer already holds an address. Adding an integer to it moves that address by **whole elements**, not by bytes — because the pointer's type tells C how large one element is.

```
uint32_t *p = &readings[0];
           // p = 0x20000100
p + 1;     // 0x20000104 (+4)
p + 2;     // 0x20000108

uint8_t *q = (uint8_t *)p;
q + 1;     // 0x20000101 (+1)
```

The compiler uses the array formula. It gets the element size from the **pointer's own type**.

$$p[i] \equiv *(p + i)$$

Array indexing *is* pointer arithmetic.

Where you will meet this

CMSIS declares the GPIO alternate-function registers as AFR[2]. These are two fixed hardware registers. But because indexing is pointer arithmetic, GPIOA->AFR[1] computes the right address by itself.

Three Sentences to Remember

So far, three ideas are enough to explain how C reaches an address of its choosing. Learn these; everything else in this chapter builds on them:

An array is a memory layout — consecutive, equally sized objects

A pointer is an address — a 32-bit number naming one location

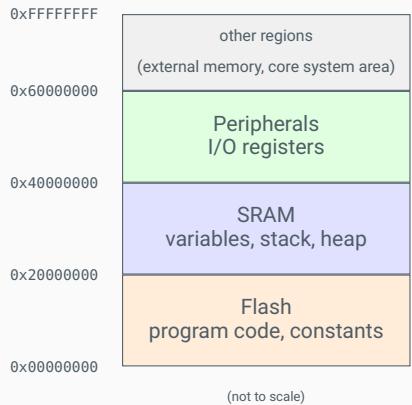
Indexing is address arithmetic — $\text{base} + (i \times \text{size})$

There is no fourth idea hiding here. A `struct`, seen later, is just a layout whose elements have different sizes. A peripheral register block, later still, is just a struct whose base address is fixed by the chip, not chosen by the compiler. Next: **where do these addresses actually lead?**

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Where Those Addresses Lead

A Cortex-M microcontroller splits its 4 GB address space into a few large regions:



Why three main kinds of block?

- **Flash** is **non-volatile**: it keeps the program after power is removed, so the chip has something to run at every reset. It reads fast but writes slowly, which fits code that rarely changes.
- **RAM** is **volatile** but fast to read *and* write: it holds the stack, heap, and variables that change constantly while the program runs. Contents are lost when power is removed.
- **Peripherals** aren't storage at all — each address is wired directly to a hardware register (GPIO, timer, UART, ...). Writing there doesn't save a value; it **changes the hardware**.

A pointer, as we just saw, is only a number. So what happens when that number points into the peripheral region?

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A peripheral register is a **hardware circuit that has been given an address**. Reading it returns the circuit's state. Writing it changes the circuit's behaviour.

We already know a pointer is just a number that identifies a location. C can reach any address through pointers. This means **C can control hardware directly** — no special language feature, no library, no operating system is needed.

```
// name the address as a pointer to a volatile 32-bit register
volatile uint32_t *gpioa_odr = (volatile uint32_t *)0x40020014;

*gpioa_odr |= (1U << 5);    // set bit 5, leave the others alone
```

That is the *whole* mechanism. The cast says: “treat this number as the address of a volatile word.” The dereference causes a load and a store. The GPIO hardware latches bit 5. Pin PA5 rises to 3.3 V.

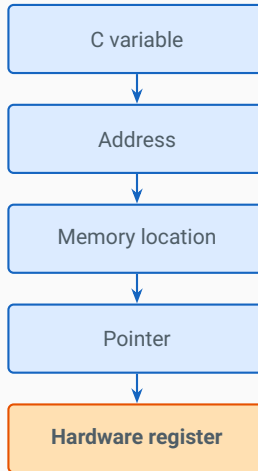
(Why `volatile`? We come back to that shortly.)

The CPU needs no special instructions for hardware: LDR/STR work everywhere, and the address decoder alone decides whether an access lands in Flash, RAM, or a peripheral register. This is **memory-mapped I/O**, and it is the reason a plain C pointer is enough to control real hardware.

Still to answer

How big is one access? And can hardware change a register *without* our code running? The next two sections answer each in turn.

This is the conceptual spine of the whole book: how an ordinary C variable becomes a way to control real hardware.



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In desktop C, `int` is a reasonable default choice. In embedded C it is a **hazard**, because the C standard does not fix its width — and every access we just wrote depends on knowing the exact width.

Type	Width	Sign	Range
<code>uint8_t</code>	8-bit	Unsigned	0 to 255
<code>int8_t</code>	8-bit	Signed	−128 to 127
<code>uint16_t</code>	16-bit	Unsigned	0 to 65,535
<code>uint32_t</code>	32-bit	Unsigned	0 to 4,294,967,295
<code>int32_t</code>	32-bit	Signed	±2,147,483,647

Two rules

Match the hardware: a 32-bit register is always `uint32_t`. A 12-bit ADC result fits in `uint16_t`.

Then save space, but watch for overflow. If two `uint8_t` values add up to more than 255, the result gets cut off. Add them into a wider type instead.

Data Sizes and Alignment

A pointer's type does not only set the step size in pointer arithmetic. It also sets how many bytes one access reads or writes, and where that access is allowed to start.

Name	Bits	Bytes	C type
Byte	8	1	uint8_t
Halfword	16	2	uint16_t
Word	32	4	uint32_t

A word starting at 0x20000000 occupies 0x20000000–0x20000003. The next word starts at 0x20000004.

Alignment

- Word access should be at a **multiple of 4**
- Halfword access at an **even** address
- Aligned accesses are generally more efficient. Misaligned ones are slower, or cause a fault

Why every STM32 register offset advances by 4

Each peripheral register is one **word**. When a datasheet shows offsets 0x00, 0x04, 0x08, this is why: each one sits 4 bytes further than the last.

When Memory Can Change Without C Changing It

Every C program relies on one basic assumption, so basic that we rarely notice it: *a variable's value changes only when your code changes it.*

1. **An ordinary variable** always obeys this rule. Nothing else can touch it between two reads.
2. **The compiler relies on this too:** it may read a value **once**, store it in a register, and reuse that register on every later access.
3. **A hardware register breaks this assumption.** It is memory-mapped, so C reaches it through an ordinary pointer – but it changes because a conversion finished, or a timer overflowed, not because of a C assignment.
4. **A variable shared with an ISR** has the same problem: to the main program, a flag the ISR sets seems to change with no matching write.
5. **volatile is the fix:** it tells the compiler to re-read this address every time, and to perform every write, in order, with none skipped.

The statement to remember

`volatile` tells the compiler a value may change **outside the current sequence of C statements**. Hardware registers are just the most common reason this happens in embedded code.

```
volatile uint32_t *status = (volatile uint32_t *)0x40011000;
while ((*status & 1) == 0) {
    // Without volatile the compiler reads the address once, caches it,
    // and this loop never terminates no matter what the hardware does.
}
```

The rule

*Every pointer to a hardware register, and every variable shared with an interrupt handler, must be **volatile**.*

Chapter 9 uses this same keyword for a different situation: an ISR talks to the main loop using nothing more than a flag.

const is a promise: the program will not change this object. The practical benefit is **placement**: a const array can live in Flash instead of SRAM, so it costs no scarce RAM.

```
const uint8_t sine_table[256] = {  
    128, 131, 134, /* ... */  
}; // lives in Flash
```

That is all there is to const. It says nothing about *when* a value can change, which is exactly the question `volatile` answers.

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What a Pointer Becomes

Every pointer operation compiles into one of exactly **two** instructions: a **load** or a **store**.

C	Instruction	What happens
<code>v = *p;</code>	<code>LDR R0, [R1]</code>	Read the word at the address in R1
<code>*p = v;</code>	<code>STR R0, [R1]</code>	Write R0 to the address in R1
<code>v = p[2];</code>	<code>LDR R0, [R1, #8]</code>	Read the word 8 bytes past R1
<code>c = *bytePtr;</code>	<code>LDRB R0, [R1]</code>	Read one <i>byte</i> , zero-filled

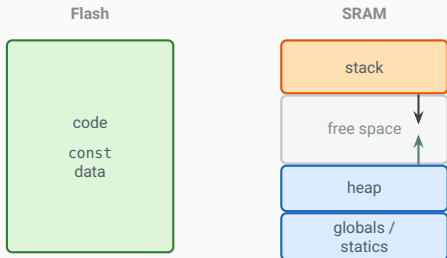
A dereference is a direction

On the **right** of an assignment, it is a **load**. On the **left**, it is a **store**. The pointer itself is just a 32-bit number in a register, supplying the address.

This is the mechanical end of the story: a pointer dereference is not an abstraction, it *is* one of these two instructions.

Where Variables Live

A pointer is an address — but who decides *which* address a variable gets? **Must understand:** the stack, global/static storage, Flash, SRAM. **Should understand:** the heap.



The stack

- Locals, arguments beyond the first four, saved registers, return addresses
- Managed automatically by **SP (R13)**
- Grows **downward** from the top of SRAM
- Chapter 7 builds an entire chapter on it

The heap

- Gives memory through `malloc()`, freed by `free()`
- **Usually avoided** in bare-metal work: it can fail at runtime when memory fragments, and its timing is not predictable
- Almost no example in this book calls `malloc`

Stack overflow — a classic embedded failure

The downward-growing stack can collide with the upward-growing global data. This **silently corrupts variables**. The usual causes are deep recursion and large local arrays.

- Memory is a numbered array of bytes. Keep the **address** separate from the **contents**.
- `&` takes an address, `*` follows one. A **pointer** is a variable that holds an address — and `p[i]` is `*(p + i)`.
- Three data sizes: **byte**, **halfword**, **word**. A word is 4 bytes — which is why register offsets advance by 4.
- A peripheral register is a circuit with an address — so **a pointer is how C touches hardware**.
- Use **fixed-width types**. Every hardware pointer must be **volatile**, since hardware can change it without your code running.
- Every dereference becomes one **LDR** or **STR** instruction.

Next: Chapter 4 — Embedded C and CMSIS

You now have the mechanism. Next, we use it to turn on a real LED.

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The brief

A colleague suggests storing a 200-sample sensor history as a global `int16_t` array, and passing it to a processing function by value, so the function "cannot corrupt the original."

Work out how much SRAM this costs at each call, out of the 128 KB the STM32F446RE provides. Explain what actually happens at the instruction level. Then design the interface you would use instead: for each parameter, say whether it should be a pointer, a const pointer, or a plain value, and why. Say which parts of the data must be declared `volatile` if an interrupt also writes to the buffer.

Questions?

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